Exp 11b: Simulation of RIP Using Cisco Packet Tracer

AIM :

To simulate the Routing Information Protocol (RIP) using the Cisco Packet Tracer simulator.

ALGORITHM :

1. Start the experiment.
2. Create the topology with three routers (Router0, Router1, Router2) connected through serial links and two PCs (PC0 and PC1) connected to Router0 and Router2.
3. Assign the IP addresses to the PCs (Desktop → IP Configuration) as given in the table.
4. Assign the IP addresses to the router interfaces from the CLI; configure the clock rate and bandwidth at the DCE end of the serial links and bring the interfaces up with no shutdown.
5. Enable the RIP routing protocol on every router using the `router rip` command.
6. Advertise the directly connected networks of each router using the `network` command.
7. Test the connectivity between PC0 and PC1 using the `ping` command and find the path using `tracert`.
8. Remove the cable between Router0 [S0/0/1] and Router2 [S0/0/1] and observe that RIP reroutes the traffic automatically through the other route.
9. Stop the experiment.

PROCEDURE :

Initial IP configuration:

Device	Interface	IP configuration	Connected with
PC0	FastEthernet	10.0.0.2/8	Router0’s Fa0/1
Router0	Fa0/1	10.0.0.1/8	PC0’s FastEthernet
Router0	S0/0/1	192.168.1.254/30	Router2’s S0/0/1
Router0	S0/0/0	192.168.1.249/30	Router1’s S0/0/0
Router1	S0/0/0	192.168.1.250/30	Router0’s S0/0/0
Router1	S0/0/1	192.168.1.246/30	Router2’s S0/0/0
Router2	S0/0/0	192.168.1.245/30	Router1’s S0/0/1
Router2	S0/0/1	192.168.1.253/30	Router0’s S0/0/1
Router2	Fa0/1	20.0.0.1/8	PC1’s FastEthernet
PC1	FastEthernet	20.0.0.2/8	Router2’s Fa0/1

Assign the IP addresses to the PCs: Double-click the PC, open the Desktop menu, click IP Configuration and enter the address from the table.

Assign the IP addresses to the router interfaces: Double-click the router, open the CLI and press Enter. Interface mode is entered from the global configuration mode. The commands used are: `interface` (enter interface mode), `ip address` (assign the address), `no shutdown` (bring the interface up) and `exit` (return to global configuration mode).

A serial interface also needs the *clock rate* and *bandwidth* parameters, which are configured only at the DCE end of the cable. The cable end can be checked with `show controllers serial 0/0/0`; if DTE is shown, these parameters are skipped.

Router0:

```
Router>enable
Router#configure terminal
Router(config)#interface fastEthernet 0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.249 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.254 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
```

Router1:

```
Router>enable
Router#configure terminal
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.250 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
```

```
Router(config-if)#ip address 192.168.1.246 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
```

Router2:

```
Router>enable
Router#configure terminal
Router(config)#interface fastEthernet 0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.245 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.253 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
```

Configure the RIP routing protocol: The routers know only their own networks and do not exchange this information on their own. RIP needs two steps: enable the RIP protocol from the global configuration mode, and tell RIP which directly connected networks to advertise.

Router0:

```
Router0(config)#router rip
Router0(config-router)#network 10.0.0.0
Router0(config-router)#network 192.168.1.252
Router0(config-router)#network 192.168.1.248
```

Router1:

```
Router1(config)#router rip
Router1(config-router)#network 192.168.1.244
Router1(config-router)#network 192.168.1.248
```

Router2:

```
Router2(config)#router rip
Router2(config-router)#network 20.0.0.0
Router2(config-router)#network 192.168.1.252
Router2(config-router)#network 192.168.1.244
```

Verification: Open the Command Prompt of PC1 and use the ping command to test the connectivity with PC0. There are two routes between PC0 and PC1:

Route 1: PC0 [10.0.0.2] ⇄ Router0 [Fa0/1 – 10.0.0.1] ⇄ Router0 [S0/0/1 – 192.168.1.254] ⇄ Router2 [S0/0/1 – 192.168.1.253] ⇄ Router2 [Fa0/1 – 20.0.0.1] ⇄ PC1 [20.0.0.2]

Route 2: PC0 [10.0.0.2] ⇄ Router0 [Fa0/1 – 10.0.0.1] ⇄ Router0 [S0/0/0 – 192.168.1.249] ⇄ Router1 [S0/0/0 – 192.168.1.250] ⇄ Router1 [S0/0/1 – 192.168.1.246] ⇄ Router2 [S0/0/0 – 192.168.1.245] ⇄ Router2 [Fa0/1 – 20.0.0.1] ⇄ PC1 [20.0.0.2]

By default RIP uses the route with the lower hop count, so Route 1 is selected; this is verified with `tracert`. To simulate the failure of Route 1, the cable between Router0 [S0/0/1] and Router2 [S0/0/1] is removed. RIP then automatically reroutes the traffic through Route 2, which can be verified by using `tracert` again.

OUTPUT :

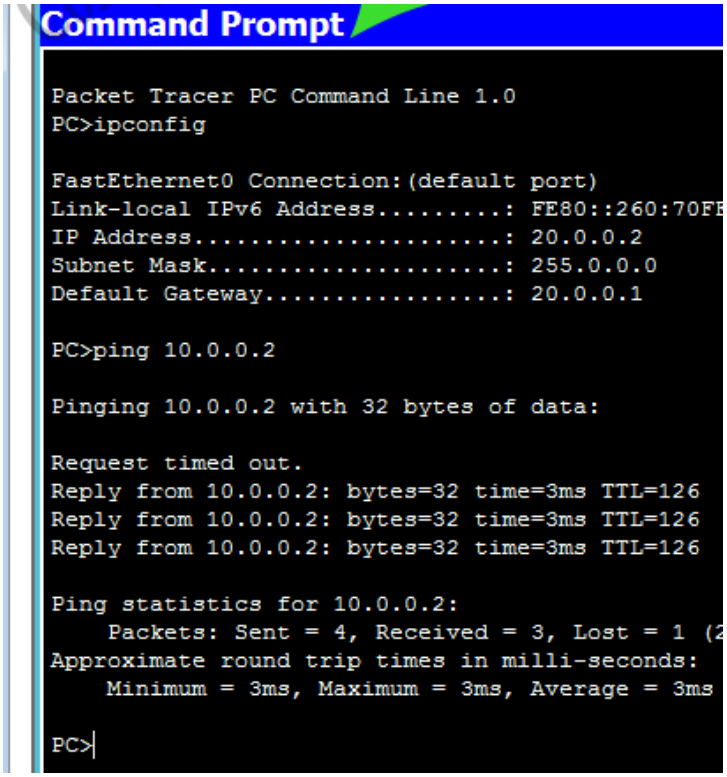


Fig 1. PC1 – IP configuration and ping to PC0 (10.0.0.2)

RESULT :

Thus, the Routing Information Protocol (RIP) was simulated using the Cisco Packet Tracer simulator. The routers exchanged their routes dynamically, and the connectivity between PC0 and PC1 was verified using the ping command.